Krishna Panthi
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Education

Clemson University | Clemson, SC | M.S. in Computer Science | GPA: 3.9/4.0 May 2026
Tribhuvan University | Kathmandu, Nepal | B.E. in Computer Engineering Apr 2021

Research Experience

Clemson University - Research Assistant | Onsite (Clemson, SC) Jan 2024 - May 2026

- Developed and optimized an open-source scientific simulation software, AquaCrop-Richards using **Python** and **NumPy** and verified the implementation using data collected in the field resulting in a peer-reviewed publication.
- Implemented deep reinforcement learning algorithms **PPO**, **DQN** and **SAC** to compare between the effects of the learning environment on optimized policies resulting in another first author paper currently under review.
- Processed and analyzed large-scale **graph** data (38M+ edges) using **Boost C++** to map upstream/downstream connectivity of US river stations across continental flowlines.
- Deployed and managed GPU nodes on an HPC cluster using **Kubernetes** for WaterSoftHack 2026.

Professional Experience

MutualArt - Software Engineer | Israel (Remote) Feb 2022 - Dec 2023

- Developed, owned and maintained full-stack applications for communication and sales serving thousands of users worldwide using **Vue.js/.NET Core** stack with **SQL Server**, **GraphQL**, **Mixpanel** and **Elasticsearch**.
- Built a marketing platform for drafting and scheduling mass-email campaigns across different continents and time zones, reaching >**200k** recipients per event and reducing campaign creation time from hours to minutes.
- Built image- and PDF-processing workflows using **Magick.NET and TextSharp** for art catalogs and invoices.
- Containerized legacy .NET applications with **Docker**, reducing deployment time.
- Used **NUnit** to write unit tests to test and validate business logic.

PensionPro - Software Engineer | Harrisburg, PA (Remote) April 2021 - Jan 2022

- Implemented custom **Angular** UI components to migrate features from a **WPF** desktop application to web application. Used Material UI and Telerik to develop dashboards and features.
- Optimized **SQL queries** and implemented **Redis** caching for faster data retrieval.

Publications

- **Panthi, K.**, Samadi, V., & Toxtli, C. (2026). A Coupled AquaCrop-Richards Model for Improved Crop Yield Prediction Through Physically Based Soil Water Dynamics. Accepted for publication at *Irrigation and Drainage*, *Wiley, 2026.*
- **Panthi, K.**, Liu, K., Samadi, V., & Toxtli, C. (2025). Deep Reinforcement Learning for Irrigation Optimization Using a Physics-Based Coupled Crop-Hydrological Model. Under review at *Computers and Electronics in Agriculture*.

Awards and Honors

- Developed a **RAG**-based document and form compliance verification system using **AWS Textract**, **Vector Database**, **AWS Bedrock**, **DynamoDB**, and **AWS Lambda**, winning Most Viable Product award at LPL Financial's 2026 Multi-University AI Hackathon (36 teams)

Skills & Interests

- **Programming Languages**: C#, JavaScript/TypeScript, Python, SQL, C++, HTML/CSS
- **Backend**: .NET Core, Flask, GraphQL, Elasticsearch, SQL Server
- **Frontend**: Vue.js, Angular, React, Mixpanel, Bootstrap, SCSS
- **Others**: Git, Docker, IIS, AWS, Azure, Kubernetes
- **AI Tools**: Claude Code, Codex, Cursor